FINAL PROJECT DOCUMENTATION

Automated Network Request Management in ServiceNow

A ServiceNow-based IT Service Management Solution

Project Title	Automated Network Request Management in ServiceNow
Platform	ServiceNow
Project Type	Academic / Practical Project
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1. Introduction

Modern organizations depend on reliable network services for daily operations. As network-related requests increase, manual handling can lead to delays, errors, and limited visibility. This project provides an automated solution in ServiceNow for submitting, processing, approving, tracking, and reporting network-related requests.

The core solution uses Service Catalog, a custom Network Database table, Flow Designer, approval processing, notifications, Incident Management, SLA Management, Reports and Dashboard, and role-based access. The final implementation was validated through structured functional testing.

2. Problem Statement

Traditional network request handling may depend on manual communication and separate tracking methods. This makes it difficult to maintain ownership, obtain approvals, communicate status, monitor resolution time, and produce consistent operational reports. The project centralizes these activities in ServiceNow and automates important stages of the lifecycle.

3. Project Objectives

- Provide a self-service mechanism for network-related requests.
- Capture request information through structured Service Catalog variables.
- Automate request processing using Flow Designer.
- Route requests for approval and handle approval outcomes.
- Maintain request status information for visibility.
- Provide Incident Management for network-related issues.
- Monitor incident resolution using SLA Management.
- Send automated notifications to relevant users.
- Provide Reports and Dashboard views.
- Apply role-based access for operational teams.
- Validate the solution through structured testing.

4. Scope of the Project

- Network Request Catalog Item and dynamic request form.
- Network Database custom table for request records.
- Approval workflow and approval-related records.
- Flow Designer automation and record updates.
- Incident creation, assignment, lifecycle, and resolution.
- Request status tracking.
- Incident SLA monitoring.
- Incident and Network Request reports.
- Dashboard visualization.
- Role-based access using groups and the itil role.
- Creation and resolution notifications.
- Functional testing.

5. Technologies and ServiceNow Components

Component	Purpose
ServiceNow	Primary ITSM platform.
Service Catalog	Self-service Network Request form.
Network Database	Custom table for storing network request information.
Flow Designer	Automates request processing, approvals, notifications, and updates.
Incident [incident]	Standard ServiceNow table for Incident Management.

Component	Purpose
SLA Management	Tracks resolution-time commitments for incidents.
Reports & Dashboard	Operational monitoring and visualization.
Groups / Roles	Controls access and separates team responsibilities.
JavaScript Client Script	Automatically sets Incident Caller to the logged-in user.

6. System Overview and Architecture

The solution separates request submission, automated processing, approval handling, incident management, SLA monitoring, and reporting while keeping the operational information inside ServiceNow.

Logical Flow

End User -> Service Catalog: Network Request -> Flow Designer -> Get Catalog Variables -> Create Network Database Record -> Approval Process -> Approved/Rejected Branch -> Notification + Record Update -> Reports & Dashboard

For network issues, the operational path is: Network Issue -> Incident -> Network Team -> SLA Monitoring -> Resolved -> Closed.

7. Project Implementation

The original implementation established the Network Request catalog item, request variables, Network Database, approval relationship, and Flow Designer automation. The final implementation extended the solution with Incident Management, request status tracking, SLA Management, reporting, dashboard visualization, role-based access, notifications, and testing.

8. Network Request Management

8.1 Service Catalog Item

A Service Catalog item named Network Request was configured under the Service Catalog. The request captures the information needed to process network services.

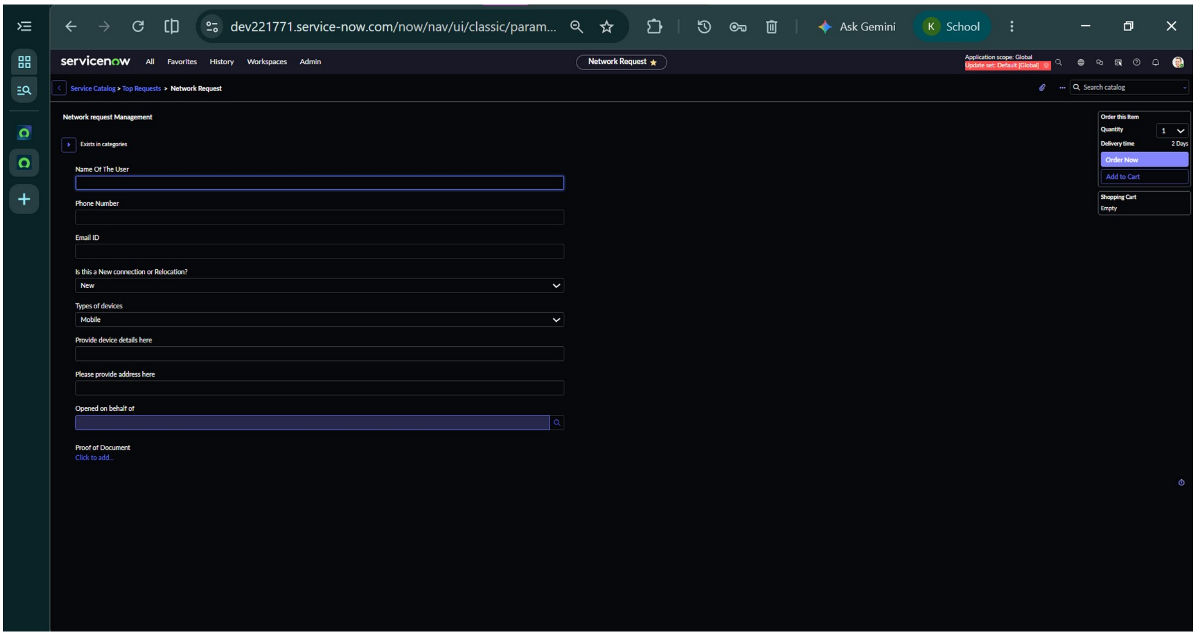

Figure 1. Network Request Catalog Item - end-user request form.

8.2 Request Variables

Configured variables include Connection Type, Relocated Address, Device Type, Address, Device Details, Other Details, Opened On Behalf Of, Email ID, User Name, Phone Number, and Status.

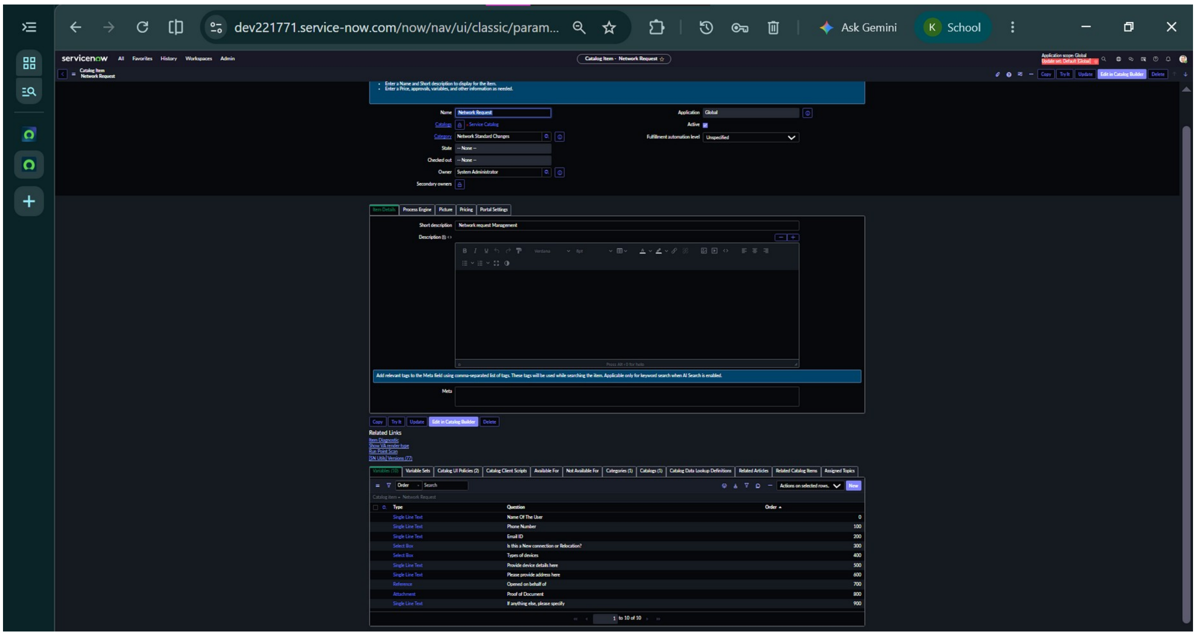

Figure 2. Network Request catalog item configuration with Variables (10).

8.3 Dynamic Form Behaviour

A UI Policy displays an additional specification field when the relevant device option is set to Others.

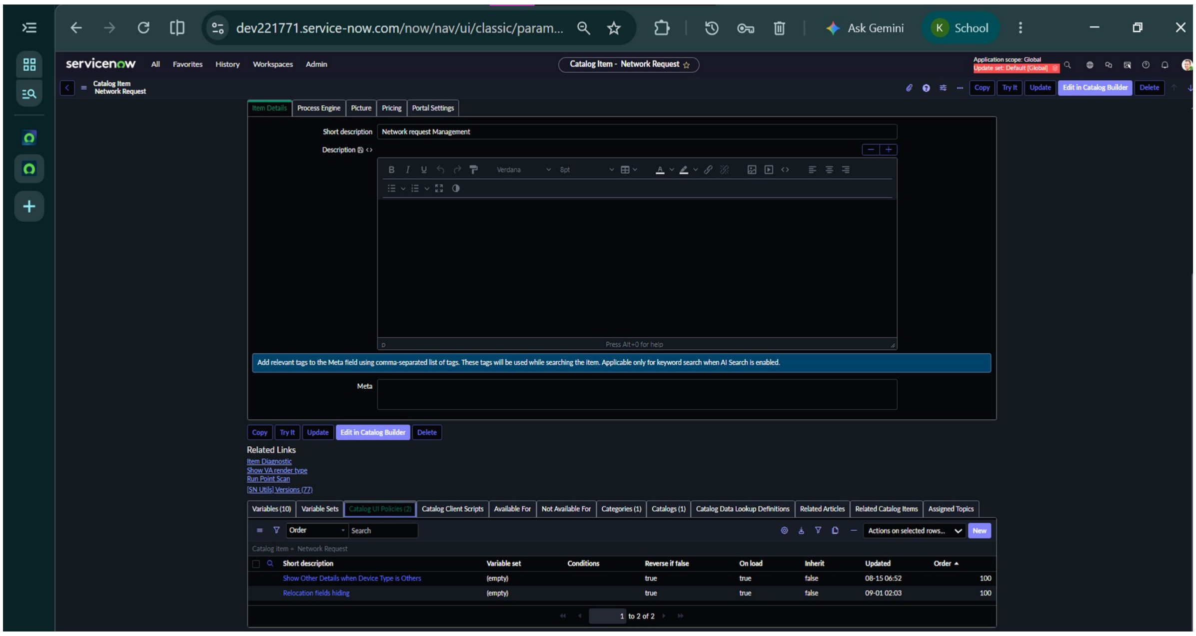

Figure 3. Catalog UI Policies configured for the Network Request item.

8.4 Network Database

The submitted request information is stored in the custom Network Database table.

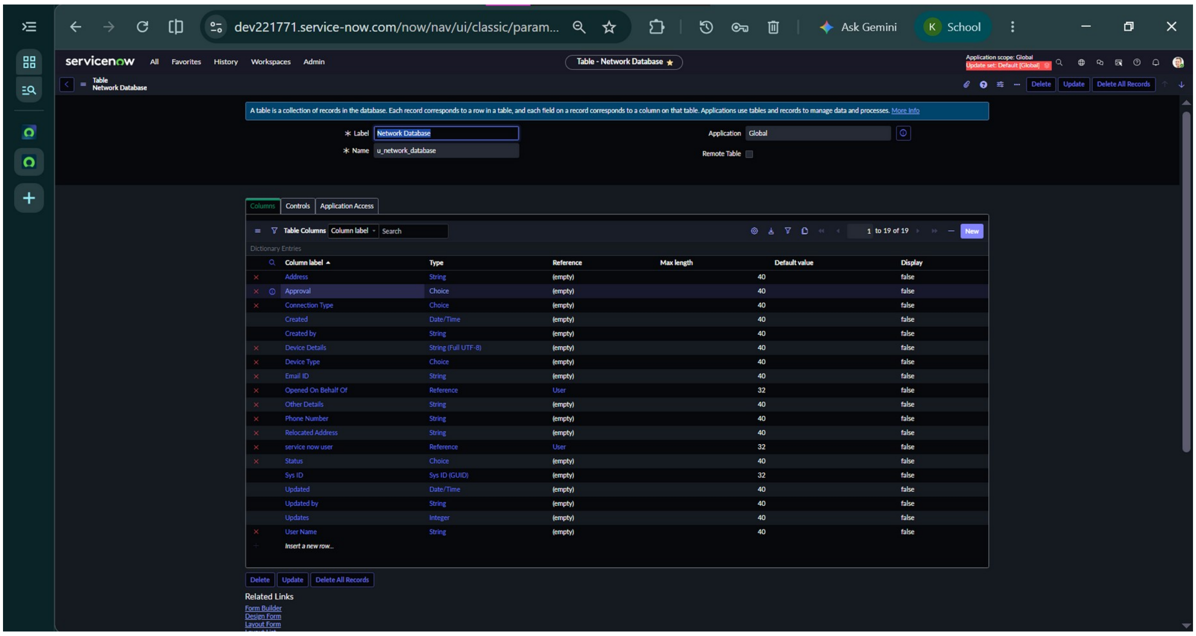

Figure 4. Network Database custom table and its column definitions.

8.5 Approval and Flow Designer

The main automation follows: Service Catalog Trigger -> Get Catalog Variables -> Create Record -> Ask for Approval -> If Approved/Rejected -> Send Email -> Update Record.

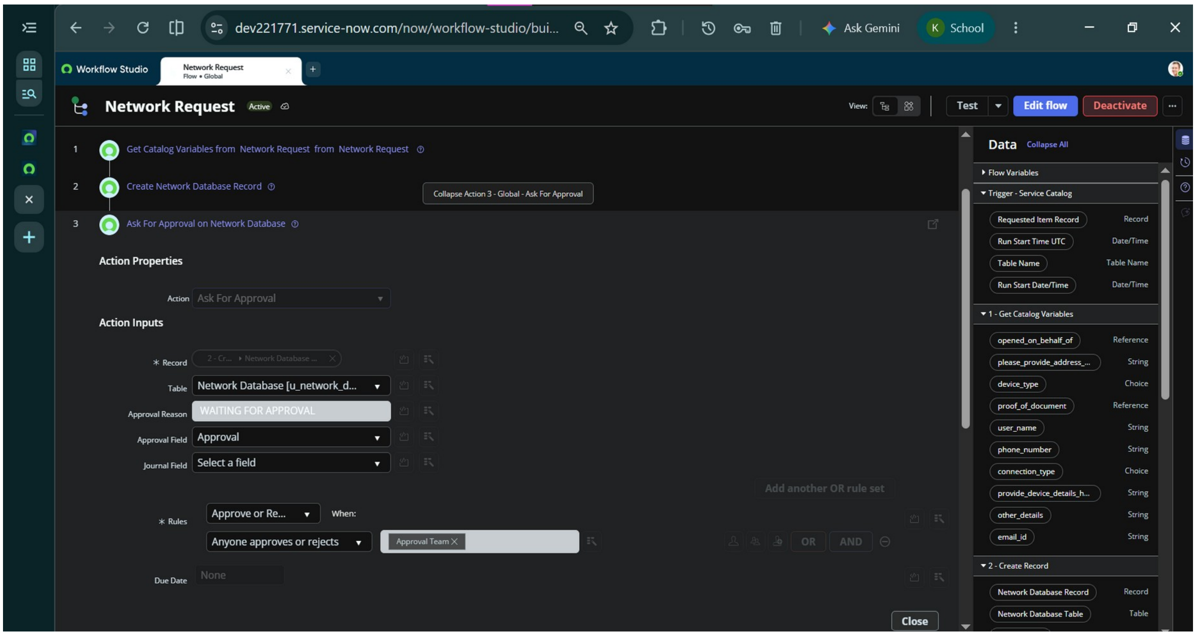

Figure 5. Ask For Approval action configuration in the Network Request flow.

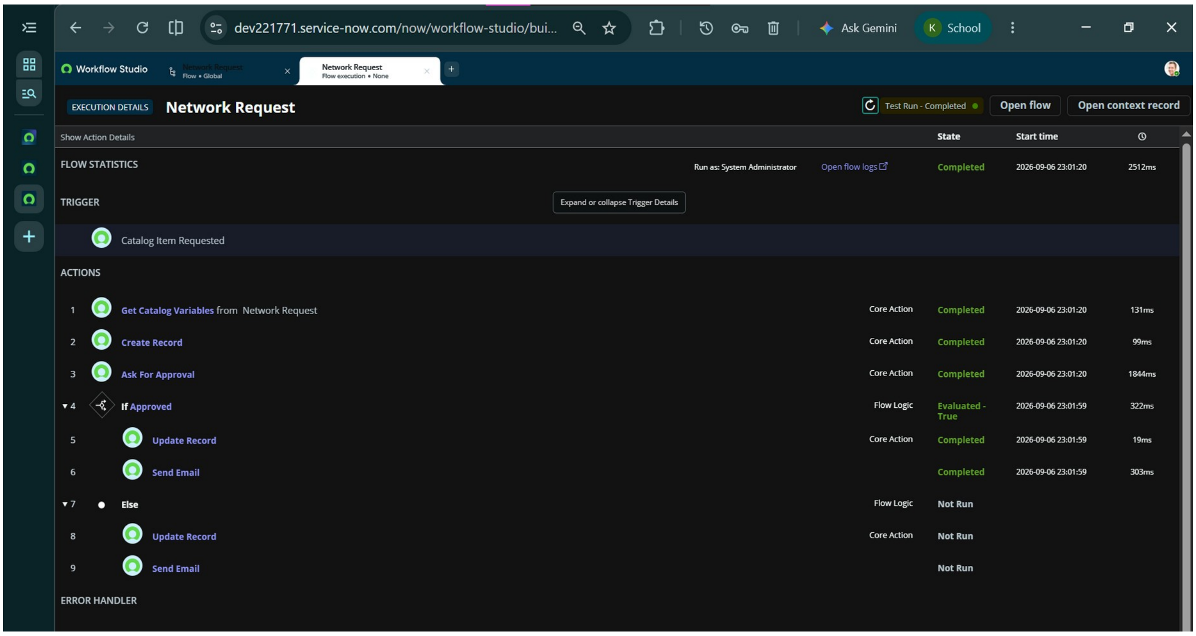

Figure 6. Network Request flow execution - request submission through approval and completion.

9. Incident Management

Incident Management uses the standard Incident [incident] table. Network incidents are assigned to the Network Team for handling.

9.1 Caller Automation

An onLoad Client Script automatically sets the Caller field to the currently logged-in user.

```
function onLoad() {     g_form.setValue('caller_id', g_user.userID); }
```

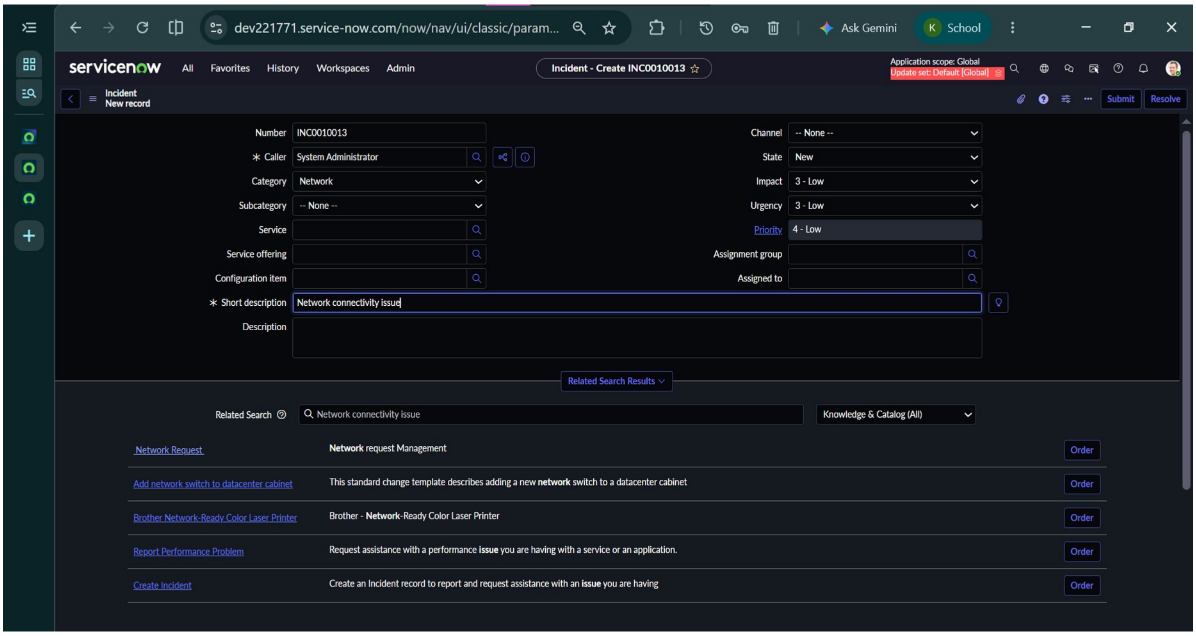

Figure 7. Incident form showing Caller automatically set for the logged-in user.

9.2 Incident Assignment

Network-category incidents are routed to the Network Team, which was configured with the itil role for ITSM access.

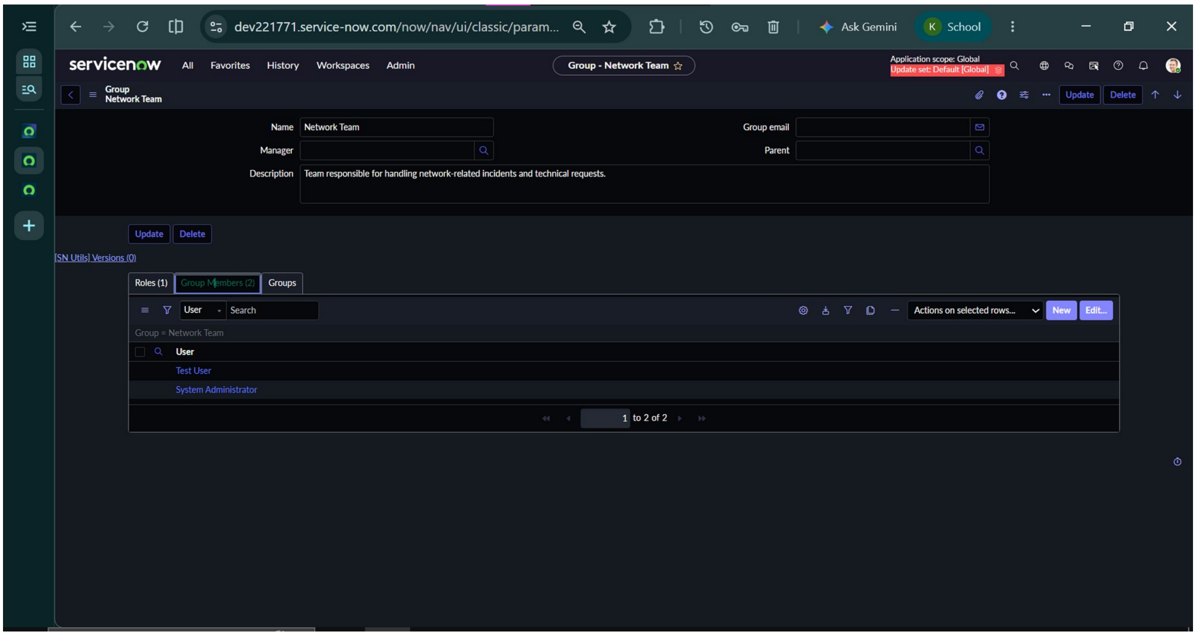

Figure 8. Network Team group configuration and members.

9.3 Incident Lifecycle

The validated lifecycle is: New -> In Progress -> Resolved -> Closed.

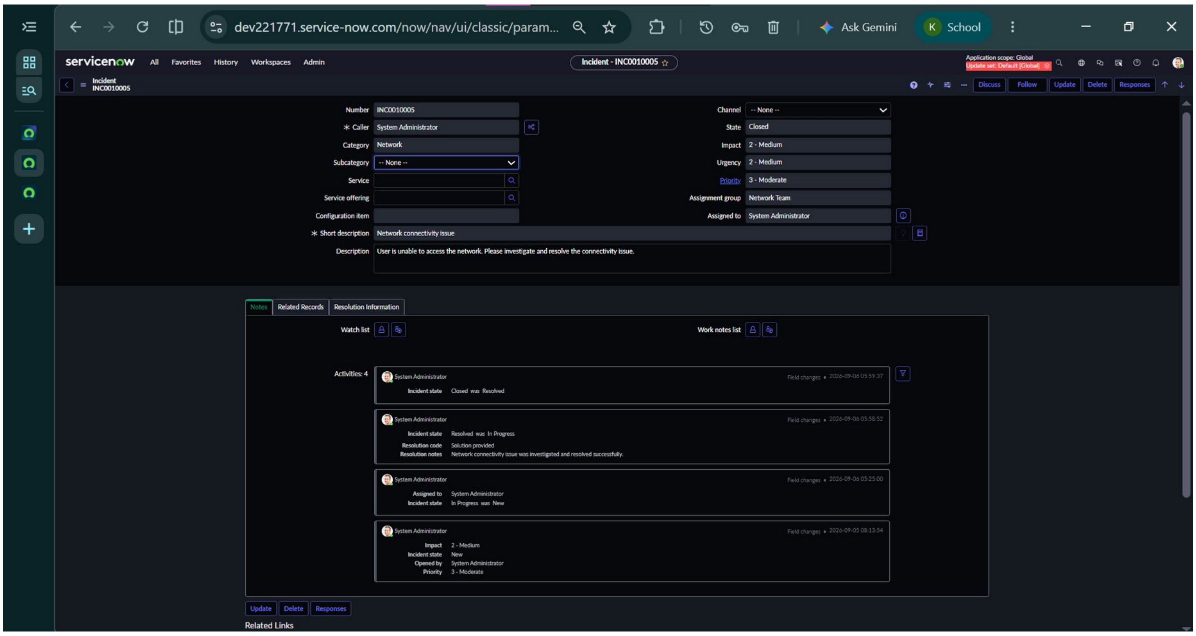

Figure 9. Completed Network Incident (INC0010005) showing assignment to Network Team and the closed lifecycle activity log.

10. Request Tracking and Status Management

The existing Status field in the Network Database table was extended to provide a clearer request lifecycle.

Status	Purpose
New	Initial or newly created record state.
Requested	Request has been submitted for processing.
In Progress	Request is actively being handled.
Approved	Approval process completed successfully.
Rejected	Request was not approved.
Completed	Request fulfillment has been completed.
Closed	Request lifecycle is closed.

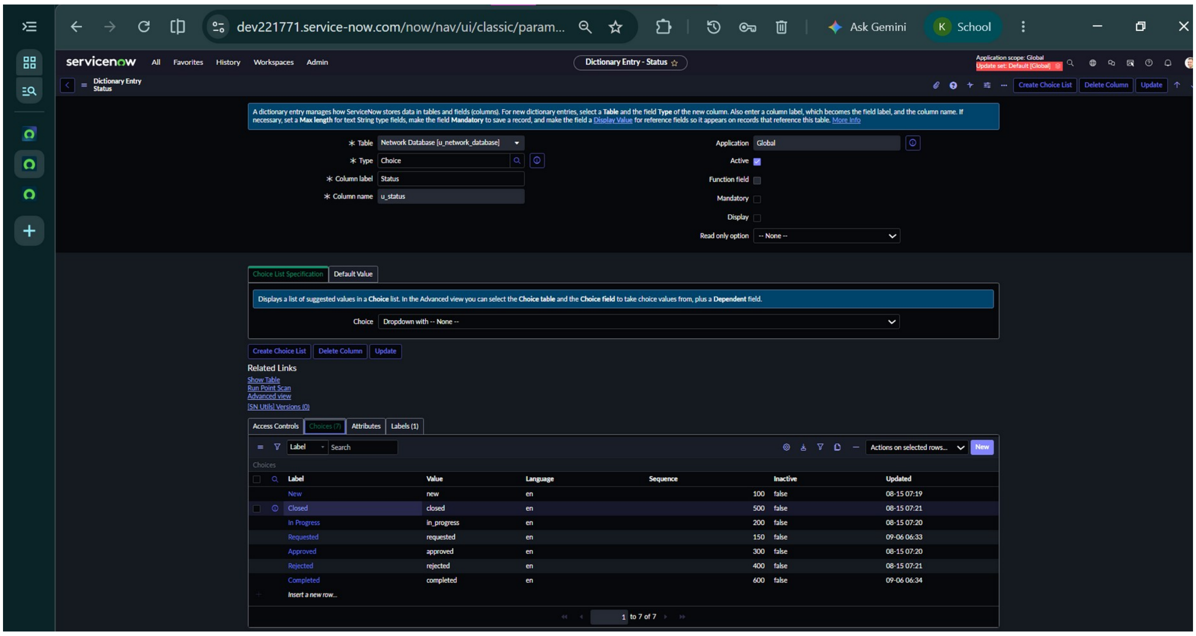

Figure 10. Status choice list configured on the Network Database table.

The report displays categories that have corresponding records. Adding a choice value does not automatically create records with that value.

11. SLA Management

SLA Management was configured for the standard Incident table using the Network Incident Resolution SLA. The demonstrated configuration uses the Incident lifecycle to start monitoring at New and stop monitoring at Resolved.

- Start: State = New
- Stop: State = Resolved

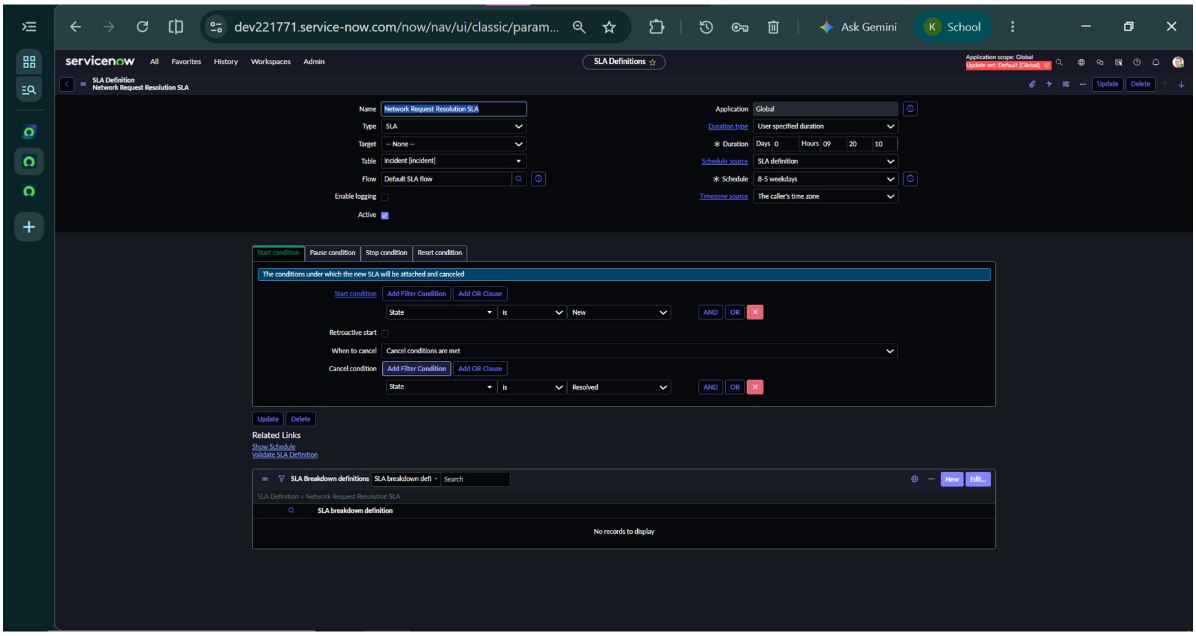

Figure 11. SLA Definition configuration for the Network Request Resolution SLA.

Note: A screenshot of the Incident Task SLAs related list was not provided for this revision; add it here once available.

12. Reports and Dashboard

Three main reports were configured and verified:

Report	Table	Purpose
Incident Status Report	Incident	Incidents grouped by State.
Incident Priority Report	Incident	Incidents grouped by Priority.
Network Request Status Report	Network Database	Requests grouped by Status.

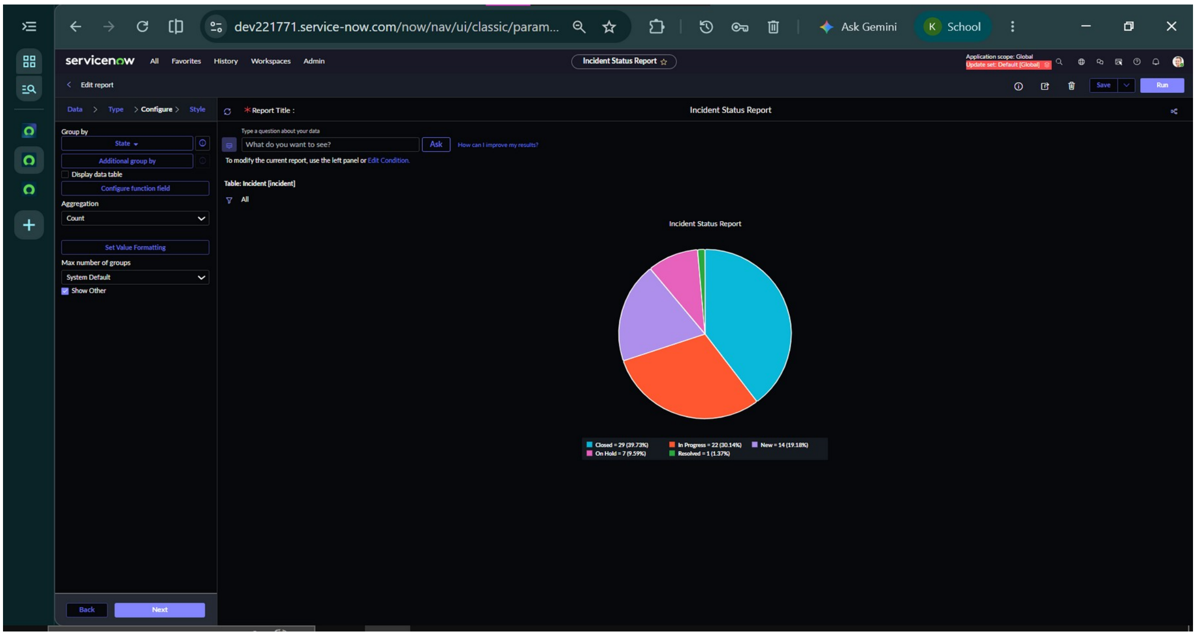

Figure 12. Incident Status Report.

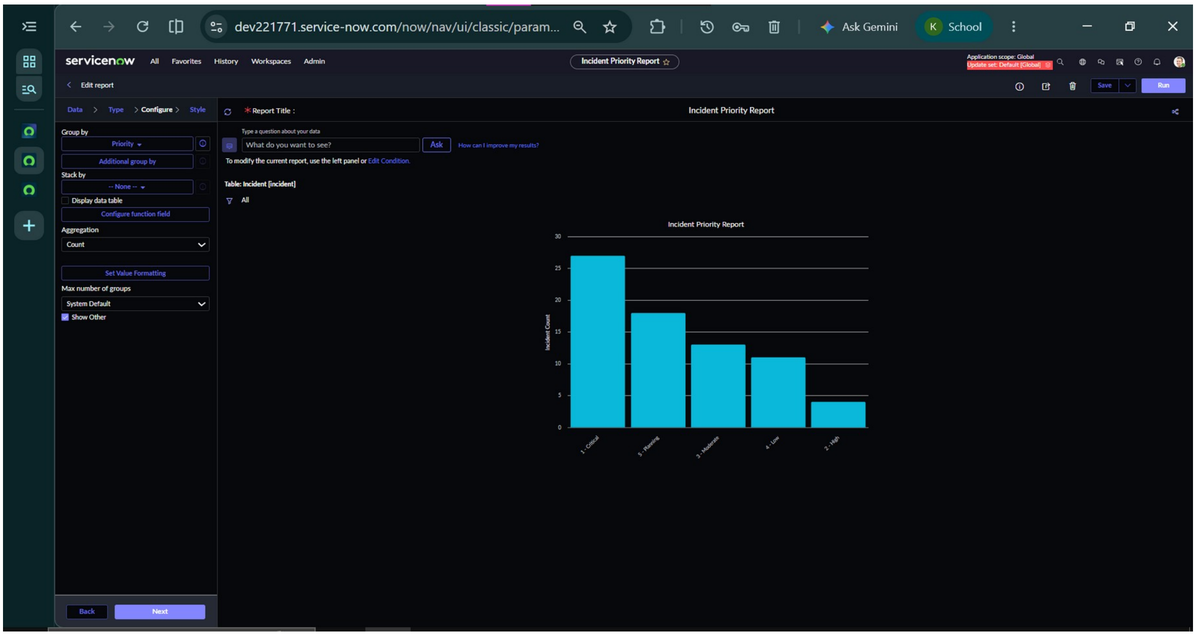

Figure 13. Incident Priority Report.

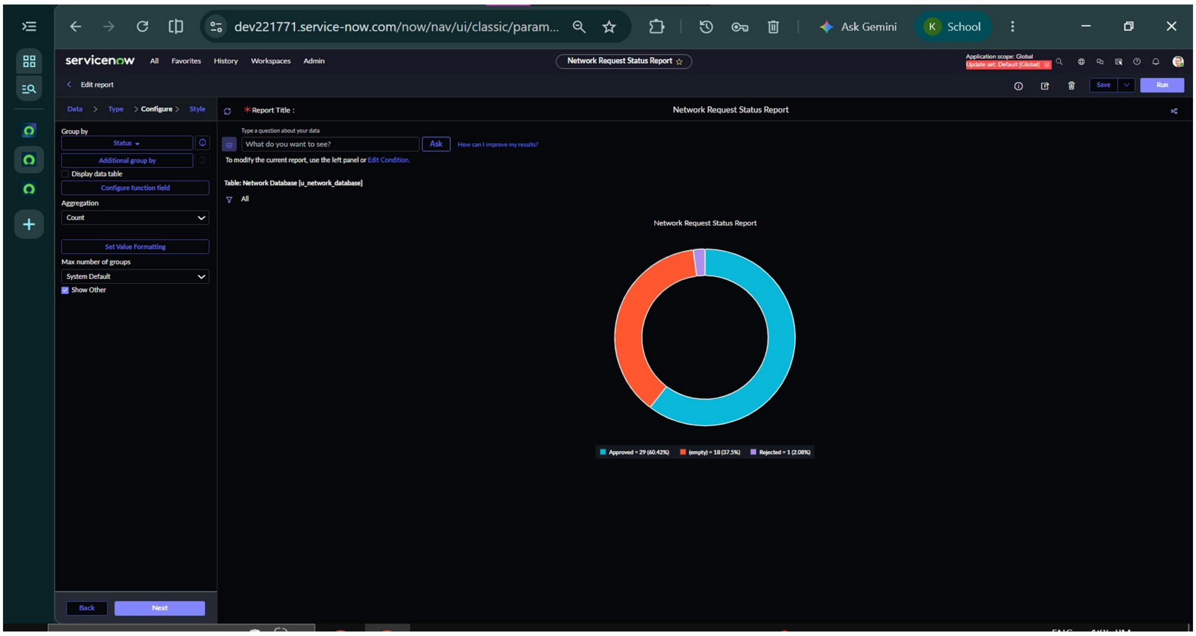

Figure 14. Network Request Status Report.

The reports were placed on the dashboard to provide a consolidated operational view of incidents and network requests.

Note: A dashboard screenshot was not provided for this revision; add it here once available.

13. Role-Based Access

Role-based access separates approval responsibilities from incident-handling responsibilities. ApprovalTeam is retained for Network Request approvals, while Network Team is used for incident handling.

The Network Team was configured with the itil role for appropriate ITSM incident access. Access was validated by testing a Network Team member's ability to access and update incidents.

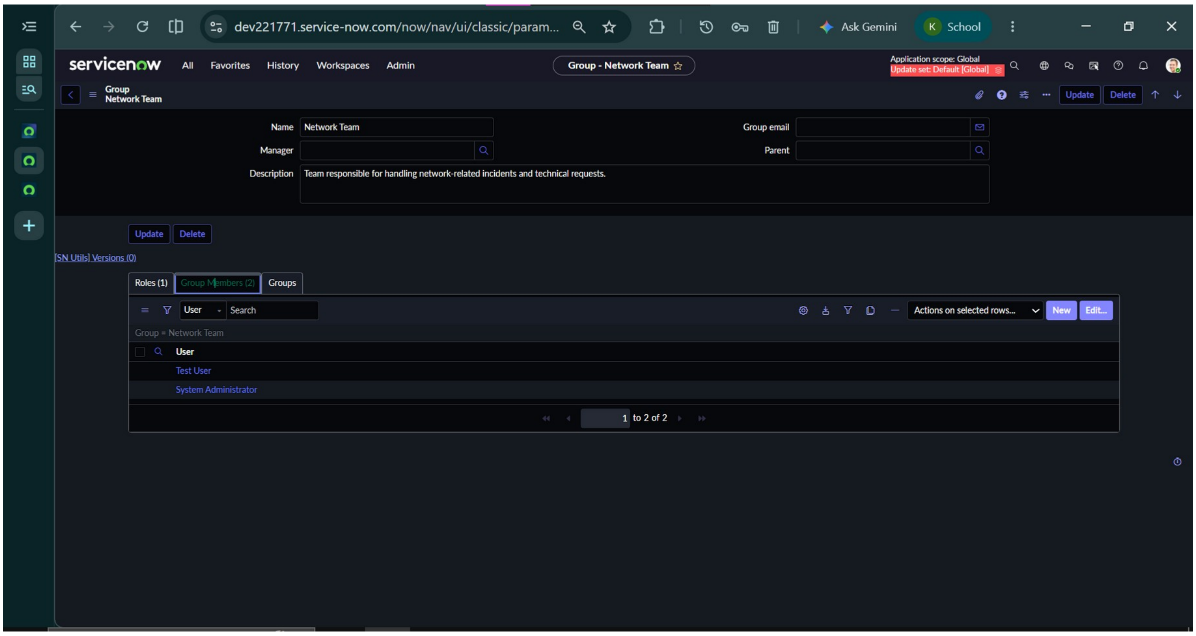

Figure 15. Network Team group, roles, and members.

14. Notifications and Automation

An Incident Management Flow uses a Record Created trigger and Network category condition to send an email to the Incident Caller. A separate Incident Resolution Notification flow uses a Record Updated trigger and Resolved state condition to notify the Caller.

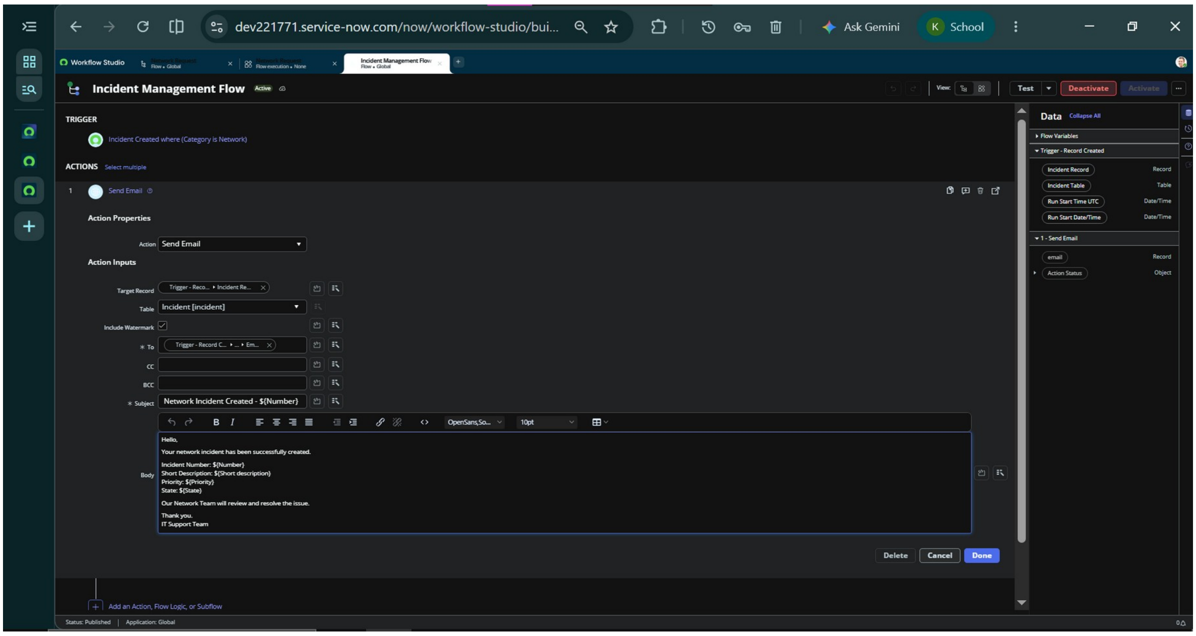

Figure 16. Incident Management Flow - creation notification email configuration.

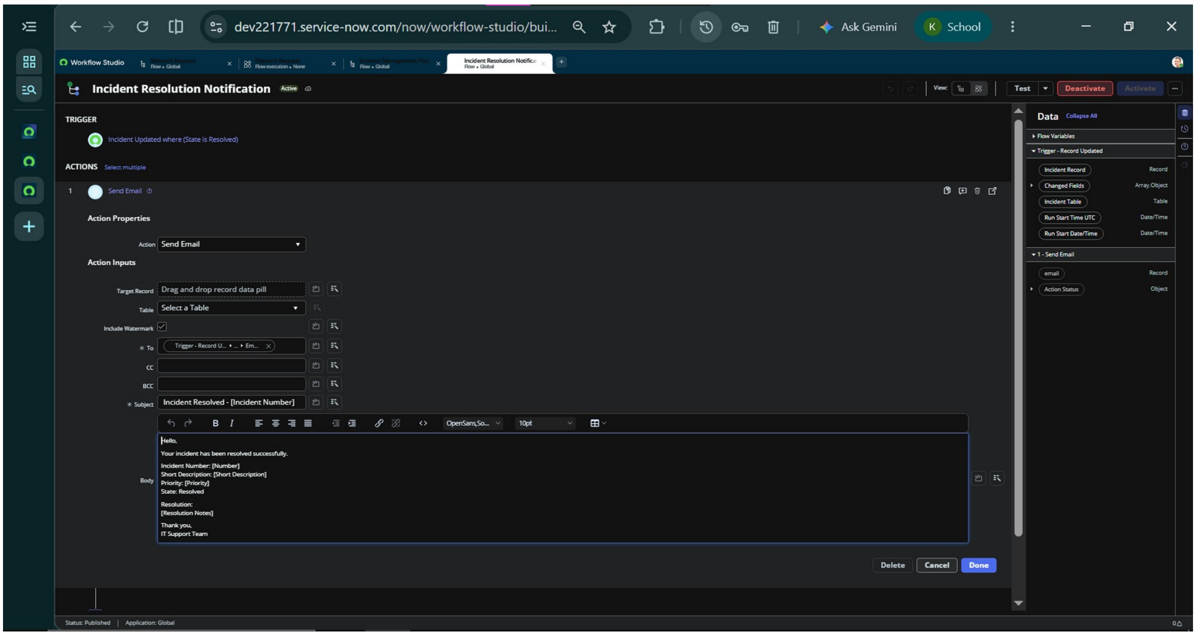

Figure 17. Incident Resolution Notification flow - resolution email configuration.

15. Testing and Validation

The major functional test cases were completed after implementation.

Test Case	Expected Result	Status
TC-01 Network Request	Request submission and configured workflow processing.	PASS
TC-02 Incident Management	New -> In Progress -> Resolved -> Closed lifecycle.	PASS
TC-03 Reports & Dashboard	Configured reports and dashboard display data.	PASS
TC-04 SLA Management	SLA attaches to a fresh incident and follows lifecycle.	PASS
TC-05 Role-Based Access	Network Team member can access and update incidents.	PASS

Testing notes: Closed incidents were not reused for earlier lifecycle testing; fresh incidents were used. Reports display status categories that have records. SLA verification was performed on a fresh incident.

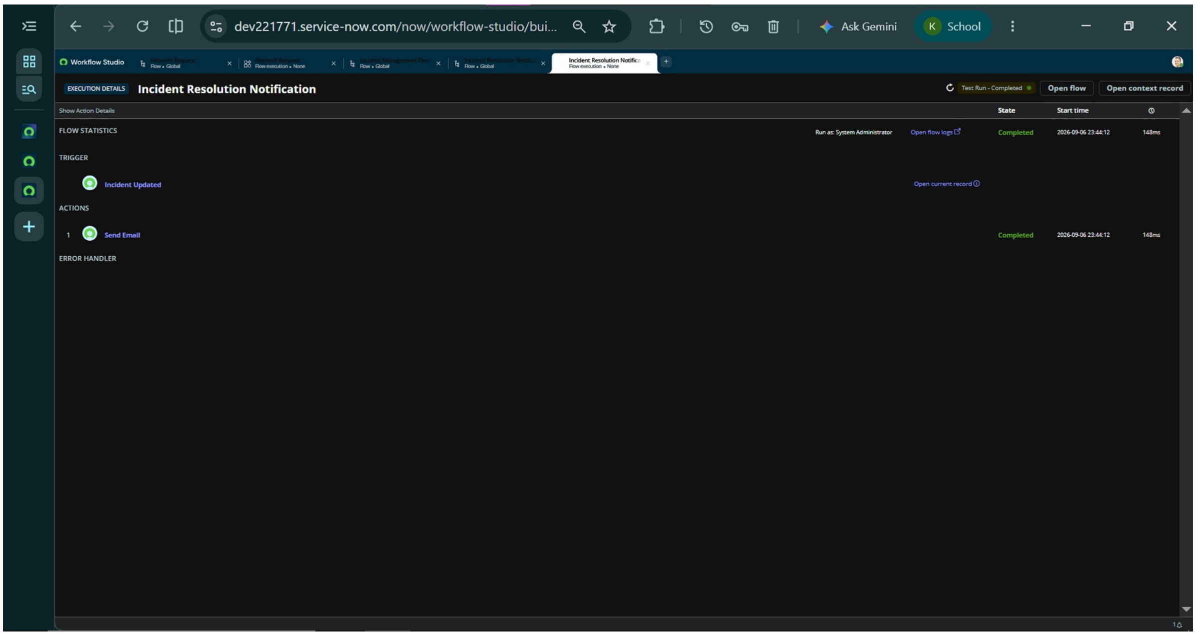

Figure 18. Incident Resolution Notification flow execution log - Test Run Completed.

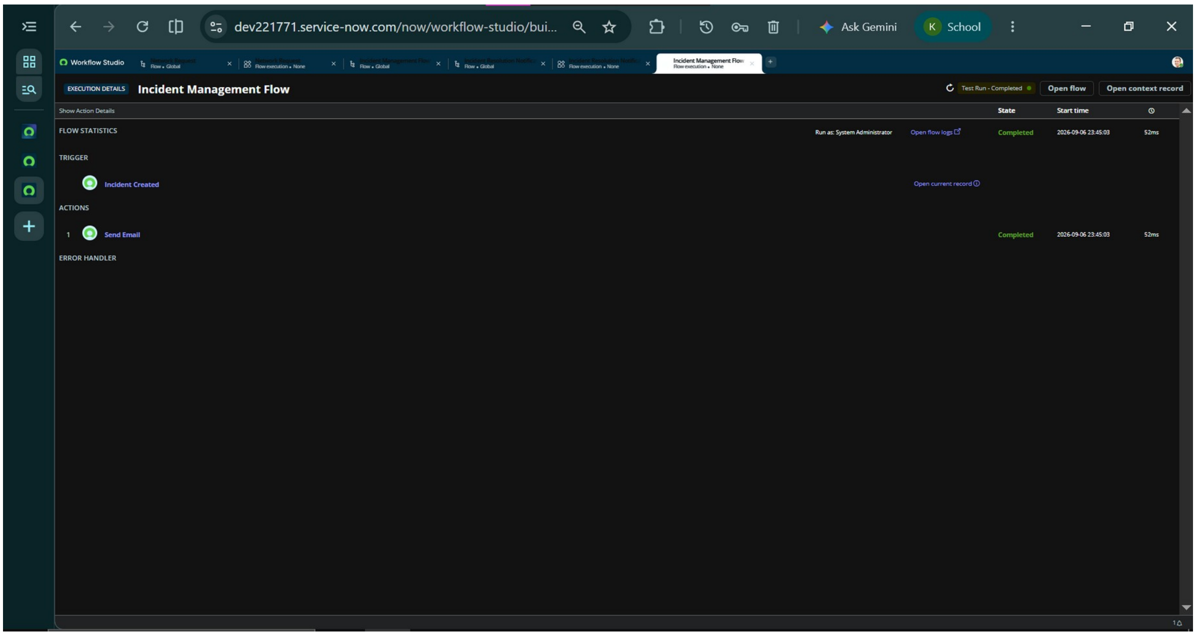

Figure 19. Incident Management Flow execution log - Test Run Completed.

16. Results and Outcomes

- Network requests can be submitted through a structured Service Catalog form.
- Request data is stored in the Network Database table.
- Approval processing is automated through Flow Designer.
- Incident Management supports assignment, lifecycle tracking, and resolution.
- Request status values provide better visibility into processing stages.
- SLA monitoring supports resolution-time tracking for network incidents.
- Reports and Dashboard provide operational visibility.
- Role-based access separates approval and incident-handling responsibilities.
- Automated notifications improve requester communication.
- The major functional test cases were completed successfully.

17. Limitations

- The demonstrated SLA configuration is implemented on the standard Incident table.
- External network automation/orchestration integration is not part of the demonstrated implementation.
- Dashboard and report values depend on records currently present in the instance.
- Production deployment would require organization-specific security, SLA, notification, and data-governance policies.

18. Future Scope

- Integrate network automation/orchestration tools for standard request fulfillment.
- Add richer approval rules based on request type, risk, or sensitivity.
- Create dedicated task-based SLA management for Network Requests.
- Add additional reports for request volume, resolution time, and SLA adherence.
- Improve portal-based requester tracking and self-service.
- Automate assignment based on request category or device type.
- Add knowledge articles and suggested solutions for common network issues.
- Integrate CMDB and Configuration Items more deeply.

19. Conclusion

The Automated Network Request Management in ServiceNow project provides a structured approach to handling network-related service requests. The solution combines Service Catalog, Flow Designer automation, approval processing, Incident Management, SLA monitoring, notifications, reporting, dashboard visualization, and role-based access.

The completed implementation reduces dependence on manual tracking by centralizing information and automating important workflow stages. It improves visibility through status tracking and reporting and provides a structured method for incident resolution and SLA monitoring.

20. Screenshot Checklist Status

#	Screenshot	Status
1	Network Request Catalog Item	Included - Figure 1
2	Network Request variables and dynamic form	Included - Figure 2
3	Network Database table and records	Included - Figure 4
4	Network Request Flow Designer	Included - Figure 6
5	Approval action configuration	Included - Figure 5
6	Approval / flow execution evidence	Included - Figure 6
7	Incident form with Caller automation	Included - Figure 7
8	Incident assigned to Network Team	Included - Figure 9
9	Incident lifecycle / resolution evidence	Included - Figure 9
10	Incident creation notification	Included - Figure 16
11	Incident resolution notification	Included - Figure 17
12	Status choices for Network Database	Included - Figure 10
13	Network Incident Resolution SLA definition	Included - Figure 11
14	Incident Task SLAs related list	Pending - not provided
15	Incident Status Report	Included - Figure 12
16	Incident Priority Report	Included - Figure 13
17	Network Request Status Report	Included - Figure 14
18	Dashboard	Pending - not provided
19	Network Team group and itil role	Included - Figure 15
20	Testing evidence	Included - Figures 18-19